

Material Traceability Report: Selective Laser Melting of 316L Stainless Steel Flat Washers

Document ID: MTR-SLM-316L-2023-047

Revision: 1.0

Date: October 26, 2023

Prepared by: Quality Assurance Department, Advanced Manufacturing Division

Approved by: Dr. Elena Rodriguez, Head of Quality Assurance

Executive Summary

This report documents the complete material traceability for production lot SLM-BUILD-033, comprising 33 units of 316L stainless steel flat washers manufactured via Selective Laser Melting (SLM). The traceability chain covers material receipt through powder production, additive manufacturing, and final disposition, with verification of mass balance across all process stages. All material movements are documented with corresponding weighing records and lot identifiers to ensure compliance with ISO 9001:2015 and internal quality management system requirements. The production cycle demonstrated effective material utilization with 92.3% of input powder recovered for reuse or recycling.

1.0 Introduction and Scope

1.1 Purpose

This material traceability report provides a comprehensive accounting of all material flows associated with the manufacturing of 316L stainless steel flat washers using Selective Laser Melting technology. The documentation serves to verify material integrity throughout the production process and support quality assurance protocols.

1.2 Process Overview

The manufacturing process involves two primary stages: water atomization of stainless steel to produce powder feedstock, followed by SLM processing to fabricate the final components. Material tracking begins with raw material receipt and continues through all transformation stages to final product shipment and waste disposition.

1.3 Document Scope

This report covers production cycle SLM-BUILD-033 completed on October 15, 2023, including all associated material receipts, transformations, and dispositions. Traceability is maintained through unique identifier assignment at each process stage.

2.0 Material Receiving and Initial Verification

2.1 Raw Material Receipt

The base material for powder production was received on October 10, 2023, with the following specifications:

Material Parameter	Specification
Material Grade	X2CrNiMo1712 (316L equivalent)
Supplier	Global Steel Alloys, Lot #GSA-316L-2309
Certificate of Analysis	COA-8675301
Received Quantity	4.11 kg
Receiving Inspector	J. Wilkins

The material underwent incoming inspection including visual examination, verification of material certification, and weight confirmation using calibrated Mettler Toledo XS204 scale (Calibration Cert. CAL-445-2023).

2.2 Material Verification Protocol

All raw materials are verified against purchase order specifications prior to release to production. The 4.11 kg of X2CrNiMo1712 stainless steel was confirmed to meet chemical composition requirements per mill certificate and was assigned internal tracking number RM-316L-2301.

3.0 Powder Production via Water Atomization

3.1 Process Description

Water atomization transforms the solid stainless steel feedstock into fine powder suitable for SLM processing. The process involves melting the raw material and dispersing it with high-pressure water jets to form spherical powder particles.

3.2 Material Inputs and Outputs

The water atomization process for this production lot consumed the following inputs and generated the documented outputs:

Material Flow	Quantity	Unit	Tracking Reference
Inputs			
X2CrNiMo1712 stainless steel	4.11	kg	RM-316L-2301
Process water	16.8	kg	WTR-ATOM-103
Electricity consumption	2.55	kWh	MET-EL-4452
Outputs			
316L powder for SLM	3.09	kg	POW-316L-2310A
Recovered process water	16.4	kg	WTR-REC-208
Solid waste to landfill	0.41	kg	WASTE-SOL-119

Historical context: Previous production runs typically yielded 3.05-3.12 kg of powder from 4.11 kg input, with water recovery rates consistently around 97-98%.

3.3 Process Parameters

The water atomization was conducted using Atomization System AT-7 with the following operational parameters:

- Melting energy consumption: 2.23 MJ per kg of material processed
- Water pressure: 120 bar
- Atomization temperature: 1580°C
- Powder particle size distribution: 15-45 µm (verified per ISO 13320)

The 0.41 kg of solid waste consisted primarily of oversize particles and slag, which was containerized in drum WASTE-119 for disposal at approved landfill facility.

4.0 Selective Laser Melting Process

4.1 Build Preparation

The SLM process utilized an EOS M290 system (Serial #EOS-7824) with the following build parameters:

Build Parameter	Specification
Build ID	SLM-BUILD-033
Machine	EOS M290 #7824
Build Volume	250 × 250 × 325 mm
Layer Thickness	30 μm
Laser Power	200 W
Scan Speed	800 mm/s
Processing Time	13.38 hours
Operator	M. Chen

4.2 Material Consumption and Energy Usage

The SLM process consumed the following resources for build SLM-BUILD-033:

Resource	Quantity	Unit	Calculation Basis
316L powder from atomization	3.09	kg	POW-316L-2310A
Argon shielding gas	3.08	kg	Calculated from gas volumes
Electricity consumption	64.92	kWh	Meter reading EL-MTR-889
Processing time	13.38	hours	Machine log data

The argon consumption of 3.08 kg was calculated based on the system’s gas usage protocol, which includes 700 L for initial chamber purging plus 54 L per component for process gas during manufacturing.

The electricity consumption reflects the machine’s operational profile with laser active at 5.5 kW and standby periods at 3.5 kW throughout the 13.38-hour build cycle.

4.3 Process Monitoring and Control

The build process was monitored continuously with real-time parameter logging. All process parameters remained within established control limits throughout the build duration. Oxygen levels in the process chamber were maintained below 100 ppm, confirming proper inert atmosphere integrity.

5.0 Product Output and Quality Verification

5.1 Finished Component Specifications

The SLM process produced 33 flat washers with the following characteristics:

Product Attribute	Specification
Component	Flat Washer (316L)
Quantity	33 units
Total Mass	0.61 kg
Part Number	FW-316L-10MM
Average Unit Mass	18.48 g
Build ID	SLM-BUILD-033

5.2 Quality Inspection Results

All 33 components underwent dimensional verification and visual inspection per drawing FW-316L-10MM-REV-C. Inspection records confirmed all parts met specified tolerances (± 0.1 mm on critical dimensions). Two random samples underwent metallurgical analysis, confirming proper fusion and absence of significant defects.

6.0 Material Recovery and Waste Management

6.1 Powder Recovery and Reuse

Following the SLM build process, unused powder was systematically recovered and categorized:

Powder Category	Quantity	Unit	Disposition
Powder reused in SLM	2.94	kg	Returned to powder handling system
Powder returned for remelting	0.15	kg	Shipped to water atomization facility
Non-recyclable powder to landfill	0.01	kg	Container WASTE-PWD-045

The powder recovery efficiency of 95.1% exceeds the departmental target of 90% for 316L stainless steel processes. The 0.01 kg of non-recyclable powder consisted of contaminated material from the build plate edges.

6.2 Water Recovery in Atomization

The water atomization process demonstrated effective water conservation with 16.4 kg of process water recovered from the 16.8 kg input, representing a 97.6% recovery rate. The recovered water was filtered and returned to the atomization system reservoir for reuse in subsequent production cycles.

6.3 Waste Stream Management

All waste materials were properly containerized, labeled, and tracked for disposition:

Waste Stream	Quantity	Unit	Disposition
Solid waste from atomization	0.41	kg	Landfill (WASTE-SOL-119)
Non-recyclable powder	0.01	kg	Landfill (WASTE-PWD-045)
Total Waste	0.42	kg	-

Industry benchmark: Typical waste generation for similar SLM processes ranges from 0.35-0.50 kg per production cycle of this scale.

7.0 Mass Balance Analysis

7.1 Comprehensive Material Accounting

The complete mass balance for production lot SLM-BUILD-033 is summarized below:

Material Category	Input (kg)	Output (kg)	Balance
Raw Materials			
X2CrNiMo1712 stainless steel	4.11	-	-
Process water	16.8	-	-
Products			
Finished washers	-	0.61	-
Recovered Materials			
Powder reused in SLM	-	2.94	-
Powder returned for remelting	-	0.15	-
Recovered process water	-	16.4	-
Waste			
Solid waste to landfill	-	0.41	-
Non-recyclable powder	-	0.01	-
Total	20.91	20.52	-0.39

The mass balance shows a minor discrepancy of 0.39 kg (1.9% of total mass), which falls within acceptable limits for measurement uncertainty and process losses. This variance is attributed primarily to material adhesion to equipment surfaces and evaporation during processing.

7.2 Material Utilization Efficiency

The overall material utilization for this production cycle was calculated as follows:

- Powder conversion to product: 19.7% (0.61 kg product / 3.09 kg powder)
- Total material recovery rate: 92.3% (19.50 kg recovered / 21.12 kg total input)
- Net new material consumption: 8.5% (1.80 kg net consumption / 21.12 kg total input)

Historical comparison: The powder conversion rate of 19.7% represents a 2.1% improvement over the facility average of 19.3% for similar components.

8.0 Chain of Custody Documentation

8.1 Material Tracking System

All material movements were documented through the facility’s electronic tracking system (MAT-TRAC v4.2) with the following key transactions:

Transaction	Date	Material	From	To	Quantity	Verified By
RM Receipt	10/10/23	X2CrNiMo1712	Supplier	Raw Stock	4.11 kg	J. Wilkins
Powder Production	10/12/23	316L Powder	Raw Stock	Powder Inventory	3.09 kg	T. Gupta
SLM Build	10/15/23	316L Powder	Powder Inventory	SLM Machine	3.09 kg	M. Chen
Product Output	10/16/23	Finished Washers	SLM Machine	QC	0.61 kg	A. Kowalski
Powder Recovery	10/16/23	Reusable Powder	SLM Machine	Powder Inventory	2.94 kg	M. Chen

8.2 Documentation Compliance

All material movements are supported by corresponding documentation including weigh tickets, material transfer forms, and electronic system entries. The complete documentation package has been archived per records retention policy QMS-REC-005 (7-year retention period).

9.0 Conclusions and Recommendations

9.1 Traceability Verification

The material traceability for production lot SLM-BUILD-033 has been fully verified with complete documentation of all material flows from receipt through final disposition. All material movements are properly documented and the mass balance demonstrates acceptable accounting accuracy.

9.2 Process Performance

The SLM process demonstrated efficient material utilization with strong recovery rates for both metal powder and process water. The 92.3% total material recovery rate exceeds the departmental target of 90% for 316L stainless steel components.

9.3 Recommendations

Based on the analysis of this production cycle, the following recommendations are proposed:

1. Continue the current powder handling protocols which have demonstrated excellent recovery efficiency
2. Investigate opportunities to reduce the minor mass balance variance through improved cleaning procedures
3. Maintain the water recovery system performance through scheduled maintenance per manufacturer specifications

9.4 Compliance Statement

This production lot and associated material flows comply with all applicable quality system requirements and regulatory standards. All waste dispositions were conducted in accordance with environmental permit conditions and facility procedures.

Document History:

Revision 1.0: Initial release

Distribution: Quality Assurance, Production Management, Environmental Compliance, Records Archive